Day 9:Creating a Docker

Objective

To learn the basics of Docker and create a Docker container for running an application in an isolated environment.

Introduction

Docker is an open-source containerization platform that enables developers to package applications along with their dependencies into lightweight containers. Containers provide a consistent runtime environment, making applications portable and easy to deploy across different systems. Docker is widely used in software development, testing, DevOps, and cloud computing.

Procedure

1. Installed and started Docker on the system.
2. Verified the Docker installation using the docker --version command.
3. Pulled a Docker image from Docker Hub using the docker pull command.
4. Created and started a Docker container using the docker run command.
5. Verified that the container was running using the docker ps command.
6. Accessed the running container using the docker exec command.
7. Tested the application inside the container.
8. Stopped the container using the docker stop command.
9. Removed the container using the docker rm command.

Commands Used

docker --version

docker pull ubuntu

docker run -it ubuntu

docker ps

docker exec -it <container_id> /bin/bash

docker stop <container_id>

docker rm <container_id>

Observations

- Docker was installed successfully.
- The required Docker image was downloaded from Docker Hub.

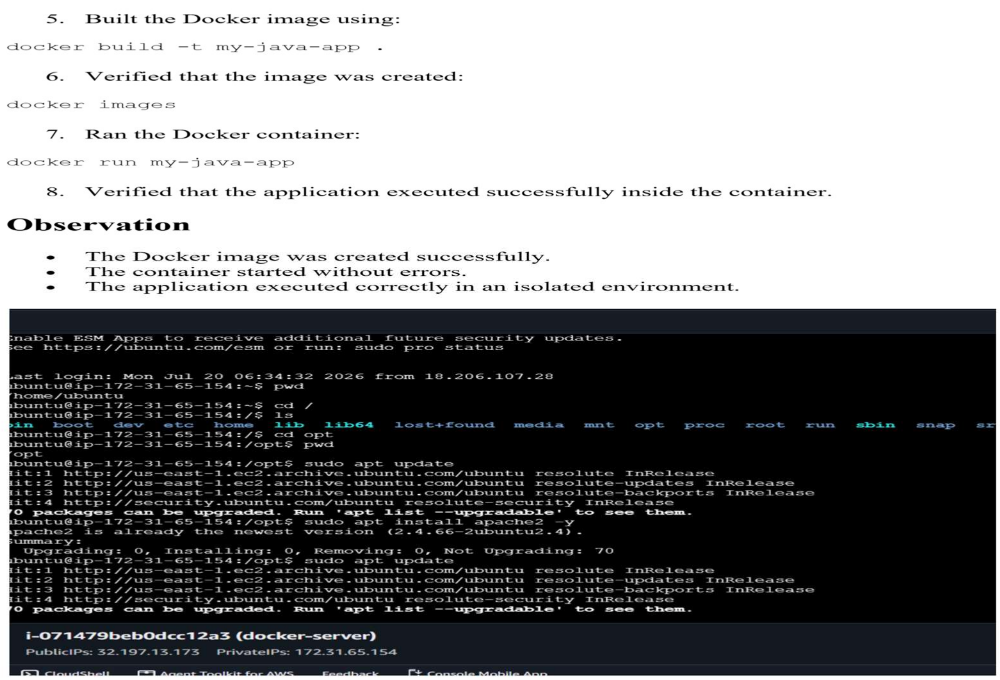
- A container was created and executed successfully.
- The container provided an isolated environment for running applications.
- The container was stopped and removed without errors.

Outcome

Successfully created, executed, managed, and removed a Docker container using basic Docker commands.

Conclusion

Docker simplifies application deployment by packaging applications and their dependencies into containers. Creating a Docker container provides a portable, consistent, and isolated environment for application development, testing, and deployment, making Docker an essential tool in modern software development and cloud computing.



5. **Built the Docker image using:**

```
docker build -t my-java-app .
```

6. **Verified that the image was created:**

```
docker images
```

7. **Ran the Docker container:**

```
docker run my-java-app
```

8. **Verified that the application executed successfully inside the container.**

Observation

- The Docker image was created successfully.
- The container started without errors.
- The application executed correctly in an isolated environment.

```
nable ESM Apps to receive additional future security updates.
ee https://ubuntu.com/esm or run: sudo pro status

ast login: Mon Jul 20 06:34:32 2026 from 18.206.107.28
buntu@ip-172-31-65-154:~$ pwd
/home/ubuntu
buntu@ip-172-31-65-154:~$ cd /
buntu@ip-172-31-65-154:/$ ls
in  boot  dev  etc  home  lib  lib64  lost+found  media  mnt  opt  proc  root  run  sbin  snap  sr
buntu@ip-172-31-65-154:/$ cd opt
buntu@ip-172-31-65-154:/opt$ pwd
/opt
buntu@ip-172-31-65-154:/opt$ sudo apt update
it:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
it:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
it:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
it:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
buntu@ip-172-31-65-154:/opt$ sudo apt install apache2 -y
pache2 is already the newest version (2.4.66-2ubuntu2.4).
ummary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 70
buntu@ip-172-31-65-154:/opt$ sudo apt update
it:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
it:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
it:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
it:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

i-071479beb0dcc12a3 (docker-server)
PublicIPs: 32.197.13.173 PrivateIPs: 172.31.65.154